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Solvency II Project

Group 01

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Text of the problem

Consider a simplified insurance company whose assets and liabilities sides are characterized as follows:

Assets

- For each policyholder there is a single fund made of equity (80%) and property (20%):
$$F_t = EQ_t + PR_t$$
- At the beginning ($t = 0$) the value of the fund is equal to the invested premium:
$$F_0 = C_0 = 100{,}000$$
- **Equity features**
  - Listed in regulated markets in the EEA
  - No dividend yield
  - Simulated with a risk-neutral GBM ($\sigma = 20\%$) and a time-varying forward rate
- **Property features**
  - Listed in regulated markets in the EEA
  - No dividend yield
  - Simulated with a risk-neutral GBM ($\sigma = 10\%$) and a time-varying forward rate

Liabilities

Contract terms

- Whole life policy
- **Benefits**
  - In case of lapse, the beneficiary gets the value of the fund at the time of lapse, with a penalty of 20 euros applied
  - In case of death, the beneficiary gets the maximum between the invested premium and the value of the fund
- **Other features**
  - Regular Deduction (RD) of 2.20%
  - Commissions to distribution channels (COMM, or trailing) of 1.40%

Model points

- Single model point representing 100 insured individuals
- Each insured is a male aged $x = 60$ at the beginning of the contract

Operating assumptions

- Mortality: rates derived from the life table SI2024
- Lapse: flat annual rate $l_t = 15\%$
- Expenses: constant unitary cost (per policy) of 50 euros per year, growing with inflation

Economic assumptions

- Risk-free rate: r derived from the yield curve (EIOPA EUR without VA at 31.12.2025)
- Inflation: flat annual rate of 2%

Other specifications

- Time step: annual
- Time horizon for the projection: 50 years
- In case of outstanding portfolio at $T = 50$, all policyholders lapse simultaneously (mass surrender) with no penalties applied
- Interest rate dynamics are deterministic, while equity and property dynamics are stochastic
- Apply the symmetric adjustment to the equity risk charge

Questions

1. Code a Matlab script to compute the Basic Solvency Capital Requirement via Standard Formula and provide comments on the results obtained. The risks to be considered are:
    - Market Interest
    - Market equity
    - Life mortality
    - Life lapse
    - Life cat
    - Expense
2. Calculate the Macaulay BEL duration in all the cases and provide comments on the results obtained.
3. Split the BEL value into its main PV components: premiums (=0), death benefits, lapse benefits, expenses, and commissions.

4. Replicate the same calculations in an Excel spreadsheet using a deterministic projection. Do the results differ from 1? If so, what is the reason behind?

5. Open questions:

    - What happens to the asset and liabilities when the risk-free rate increases/decreases with a parallel shift of, say, 100bps? Describe the effects for all the BEL components.
    - What happens to the liabilities if the insured age increases? What if there were two model points, one male and one female?

Summary table of the results

Results	MVA	BEL	d_MVA abs	d_BEL abs	d_MVA rel	d_BEL rel	BoF	d_BoF	dur_L
BASE	10000000	9655842.76	-	-	-	-	344157.24	-	5.68346226
IR_up	10000000	9630314.31	0	25528.45	0	-0.00264	369685.69	-25528.45	5.66255972
IR_dw	10000000	9687660.06	0	31817.30	0	0.00329	312339.94	31817.30	5.71109769
EQ	6248000	6197571.47	3752000	3458271.29	-0.3752	-0.35807	50132.19	294025.05	5.73199708
PROP	9500000	9196612.70	500000	459230.06	-0.05	-0.04756	303387.30	40769.94	5.68724199
MORT	10000000	9668897.88	0	13055.12	0	0.00135	331102.12	13055.12	5.64822929
CAT	10000000	9653146.80	0	2695.96	0	-0.00028	346853.20	-2695.96	5.67399196
LAPSE UP	10000000	9738765.68	0	82922.92	0	0.00859	261234.32	82922.92	4.04124761
LAPSE DOWN	10000000	9546521.58	0	109321.18	0	-0.01132	453478.42	-109321.18	9.31091608
LAPSE MASS	10000000	9742905.58	0	87062.82	0	0.00901	257094.42	87062.82	4.28111711
EXPENSES	10000000	9661180.26	0	5337.50	0	0.00055	338819.74	5337.50	5.68194406
BSCR	376360	-	-	-	-	-	-	-	-

Table 1.1: Stochastic Calculation Results

Specification of the Formulas

The valuation has been performed under a risk-neutral framework, consistently with Solvency II principles.

Discount factors are used to compute the present value of future cash flows, reflecting the time value of money.

$$DF_t = \frac{1}{(1 + r_t)^t}$$

Forward Rates represent the instantaneous growth rate of assets under the risk-neutral measure.

$$f_t = \log\left(\frac{DF_{t-1}}{DF_t}\right)$$

Asset Dynamics: The fund evolves stochastically, with equity representing the main source of volatility.

$$F_0 = F_0^{EQ} + F_0^{PR}$$

$$F_1^{EQ} = F_0^{EQ} \cdot \exp\left(f_0 - \frac{1}{2}\sigma_{EQ}^2 + \sigma_{EQ} Z_0\right)$$

$$F_1^{PR} = F_0^{PR} \cdot \exp\left(f_0 - \frac{1}{2}\sigma_{PR}^2 + \sigma_{PR} Z_0\right)$$

Net Fund Evolution: At each time step, the fund is reduced by regular deductions and the next period evolution starts from this net value.

$$F_t^{net} = F_t \cdot (1 - RD)$$

$$F_{t+1}^{EQ} = F_t^{EQ} \cdot (1 - RD) \cdot \exp\left(f_{t+1} - \frac{1}{2}\sigma_{EQ}^2 + \sigma_{EQ} Z_{t+1}\right)$$

$$F_{t+1}^{PR} = F_t^{PR} \cdot (1 - RD) \cdot \exp\left(f_{t+1} - \frac{1}{2}\sigma_{PR}^2 + \sigma_{PR} Z_{t+1}\right)$$

This reflects the fact that the regular deduction is applied before the next period evolution.

Policyholder Evolution: The portfolio consists of 100 policyholders, whose evolution is modeled at the individual level within each simulation. At the beginning, all policyholders are active. At each time step, death and lapse events are simulated independently for each individual using Bernoulli trials based on the corresponding mortality and lapse probabilities. For each individual still active, a random variable uniformly distributed in $(0, 1)$ is generated: if its realization is lower than the death probability, the policyholder is considered dead and removed from the active population; conditional on survival, the same procedure is then applied to lapse events, using the lapse probability instead of the

death probability. As a result, the number of active policyholders N_t follows a stochastic decreasing process and varies across simulations around its expected trajectory.

Cash Flows arise from contractual guarantees, operating costs and profit margins embedded in the product.

Death benefits:

$$CF_t^{death} = \max(F_t^{net}, C_0) \cdot N_t^{death}$$

Lapse benefits:

$$CF_t^{lapse} = (F_t^{net} - 20) \cdot N_t^{lapse}$$

Expenses:

$$CF_t^{exp} = e_0 \cdot (1 + \text{inflation})^t \cdot N_t$$

Expenses are computed at the end of each period, applied only to the number of surviving policyholders and increased over time according to the inflation rate. The resulting cash flows are then discounted to present value.

Commissions:

$$CF_t^{comm} = 1.40\% \cdot F_t^{gross} \cdot N_t$$

Commissions are calculated at the end of each period on the fund value before deductions and are proportional to the number of active policyholders. These cash flows are also discounted.

Regular deductions:

$$CF_t^{RD} = 2.20\% \cdot F_t^{gross} \cdot N_t$$

Regular deductions represent a proportional charge applied to the fund and constitute the main source of profit for the insurer. They are applied before the evolution of the fund in the following period and therefore directly affect the future asset dynamics.

Best Estimate Liabilities: The BEL represents the expected present value of future obligations under the risk-neutral measure.

$$BEL = \mathbb{E}^{\mathbb{Q}} \left[\sum_{t=1}^{T} DF_t \cdot (CF_t^{death} + CF_t^{lapse} + CF_t^{exp} + CF_t^{comm}) \right]$$

Basic Own Funds: measure the solvency position of the insurer.

$$BoF = MVA - BEL$$

Solvency Capital Requirement: measures the loss in Own Funds under stress scenarios.

$$SCR_i = \max(BoF_{base} - BoF_{shock}, 0)$$

Aggregation of the SCR components is performed using correlation matrices, consistently with the Solvency II Standard Formula, in particular:

Life Risk Aggregation: The Life risks exhibit partial dependence, with stronger links between lapse and expenses, reflecting behavioural and operational interactions, and its SCR computed as:

$$SCR_{life} = \sqrt{SCR'_{life} \cdot \rho_{life} \cdot SCR_{life}}$$

with:

$$SCR_{life} = \begin{pmatrix} SCR_{mort} \\ SCR_{lapse} \\ SCR_{expense} \\ SCR_{cat} \end{pmatrix}$$

and correlation matrix:

$$\rho_{life} = \begin{pmatrix} 1 & 0 & 0.25 & 0.25 \\ 0 & 1 & 0.5 & 0.25 \\ 0.25 & 0.5 & 1 & 0.25 \\ 0.25 & 0.25 & 0.25 & 1 \end{pmatrix}$$

Market Risk Aggregation: The Market risks show strong dependence between equity and property, while the correlation with interest rates depends on the direction of the stress scenario. Its SCR is computed as:

$$SCR_{mkt} = \sqrt{SCR'_{mkt} \cdot \rho_{mkt} \cdot SCR_{mkt}}$$

with:

$$SCR_{mkt} = \begin{pmatrix} SCR_{IR} \\ SCR_{EQ} \\ SCR_{PR} \end{pmatrix}$$

The correlation matrix depends on the dominating interest rate scenario:

If the downward shock dominates:

$$\rho_{mkt} = \begin{pmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.75 \\ 0.5 & 0.75 & 1 \end{pmatrix}$$

Otherwise:

$$\rho_{mkt} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0.75 \\ 0 & 0.75 & 1 \end{pmatrix}$$

BSCR Aggregation: The aggregation between life and market risks reflects diversification effects, reducing the overall capital requirement compared to a simple sum of risks. The final BSCR is computed as:

$$BSCR = \sqrt{SCR_{life}^2 + SCR_{mkt}^2 + 2 \cdot \rho \cdot SCR_{life} \cdot SCR_{mkt}}$$

with:

$$\rho = 0.25$$

Stress Scenarios

1a Interest Rate Shock

Interest rate shocks are directly taken from the EIOPA shocked yield curves (without volatility adjustment):

$$DF^{up}(t) = \frac{1}{(1+r_t^{up})^t}, \quad DF^{down}(t) = \frac{1}{(1+r_t^{down})^t}$$

$$f_t^{up} = \log\left(\frac{DF^{up}(t-1)}{DF^{up}(t)}\right), \quad f_t^{down} = \log\left(\frac{DF^{down}(t-1)}{DF^{down}(t)}\right)$$

The impact of interest rate movements is driven by the discounting mechanism applied to future cash flows. Higher rates reduce the present value of liabilities, leading to an increase in BoF, while lower rates produce the opposite effect. The relatively limited variation observed reflects the medium duration of the liability profile, which moderates the sensitivity to yield curve shifts.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{up} = 369{,}686 \quad BoF_{down} = 312{,}340}$$

In this case the SCR is defined as:

$$SCR_{ir} = \max(SCR_{IR_{up}}, SCR_{IR_{down}})$$

1b Equity Shock

Equity is shocked according to Solvency II Type 1 shock:

$$\alpha_{EQ} = 39\% + SA$$

with $SA = 0.079$ according to EIOPA

$$F_0^{EQ} = 0.8 \cdot F_0 \cdot (1 - \alpha_{EQ})$$

The equity shock strongly reduces the value of the fund, which directly impacts the asset side. Although liabilities also decrease due to lower expected fund values, the presence of guarantees prevents a proportional reduction. This asymmetry leads to a significant erosion of Own Funds.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{EQ} = 50{,}132 \quad \Delta BoF_{EQ} = 294{,}025}$$

1c Property Shock

$$\alpha_{PR} = 25\%$$

$$F_0^{PR} = 0.2 \cdot F_0 \cdot (1 - \alpha_{PR})$$

The effect of the property shock is more contained due to both the lower exposure and the smaller stress applied. The resulting decrease in BoF is consistent with a proportional but limited impact on the overall fund value.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{PR} = 303{,}387 \quad \Delta BoF_{PR} = 40{,}770}$$

1d Mortality Shock

$$q_x^{shock} = \min((1 + 15\%) \cdot q_x, 1)$$

An increase in mortality accelerates the occurrence of death benefits, shifting cash flows to earlier periods and increasing their present value. This results in a higher BEL and a reduction in BoF.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{mort} = 331{,}102 \quad \Delta BoF_{mort} = 13{,}055}$$

1e Catastrophe Shock

$$q_{x_0}^{shock} = q_{x_0} + 0.0015$$

The catastrophe shock has a negligible impact on the portfolio, as it affects only a small fraction of policies. The slight variation observed is not economically meaningful and could be mainly driven by simulation noise.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{cat} = 346{,}853 \quad \Delta BoF_{cat} = -2{,}696}$$

1f Lapse Shock

$$l^{up} = \min((1 + 50\%) \cdot l, 1)$$

$$l^{down} = 50\% \cdot l$$

$$l^{mass} = 40\%$$

The mass lapse scenario assumes that at the first time step 40% of the policyholders surrender the contract, instead of the base lapse rate of 15%. From the second year onward, the lapse rate returns to its base level. This shock captures a sudden and simultaneous surrender behaviour across the portfolio.

Lapse behaviour significantly affects both the timing of cash flows and the generation of future profits. Higher lapse rates lead to earlier surrenders, increasing immediate outflows and reducing the number of policies that remain in force, which in turn increases the BEL and reduces BoF. On the contrary, lower lapse rates allow the insurer to retain policies for longer, enhancing the contribution of future deductions and improving solvency.

In line with the Solvency II Standard Formula, the lapse SCR is determined by selecting the most adverse scenario, i.e. the one leading to the largest loss in BoF compared to the base case.

$$SCR_{lapse} = \max(SCR_{lapse_{down}}, SCR_{lapse_{up}}, SCR_{lapse_{mass}})$$

In our case, the mass lapse scenario produces the highest deterioration and is therefore used for the SCR calculation.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{up} = 261{,}234 \quad BoF_{down} = 453{,}478 \quad BoF_{mass} = 257{,}094}$$

$$\boxed{\Delta BoF_{up} = 82{,}923 \quad \Delta BoF_{down} = -109{,}321 \quad \Delta BoF_{mass} = 87{,}063}$$

1g Expense Shock

$$e_0^{shock} = e_0 \cdot (1 + 10\%)$$

$$inf^{shock} = inf + 1\%$$

Higher expenses directly increase the cost component of liabilities, leading to a higher BEL and a reduction in BoF. The effect remains limited compared to market and lapse risks.

$$\boxed{BoF_{base} = 344{,}157 \quad BoF_{exp} = 338{,}820}$$

$$\boxed{\Delta BoF_{exp} = 5{,}337}$$

1h Number of Scenarios and Martingale Test

A total of **250,000** Monte Carlo scenarios have been simulated. The adequacy of the number of scenarios has been assessed by analysing the statistical precision of the estimated quantities, in particular the Best Estimate Liabilities (BEL) and the Basic Own

Funds (BoF).

Since the BoF is defined as:

$$BoF = MVA - BEL$$

and the Market Value of Assets (MVA) is deterministic, the confidence interval of the BoF has the same absolute width as that of the BEL.

However, the magnitude of the BoF is significantly smaller than that of the BEL. As a consequence, a confidence interval that appears sufficiently tight in absolute terms for the BEL may correspond to a much larger relative error when referred to the BoF.

This implies that the statistical precision of the BoF estimate is effectively lower, even though the absolute uncertainty remains unchanged. Therefore, a larger number of Monte Carlo simulations is required to achieve a satisfactory level of relative accuracy for the BoF.

With the selected number of simulations, we performed a martingale test on the evolution of the fund at each time step. This consists in verifying that the discounted fund value, normalized by the initial notional, has expectation equal to 1 under the risk-neutral measure, i.e.

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{F_t}{F_0} \cdot DF_t\right] = 1. \forall t \in [0, T]$$

In our case, the test yields values as shown in the following plot, confirming the consistency of the simulation.

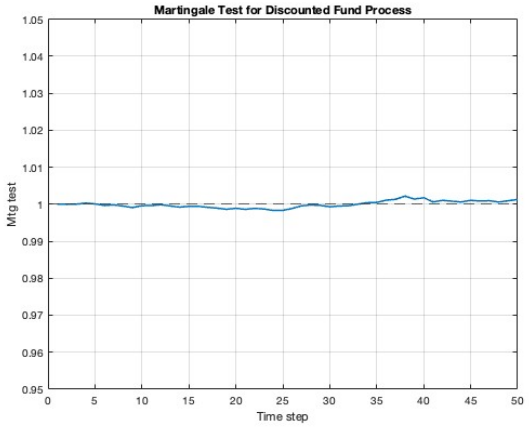


Figure 1.1: Martingale Test

Deterministic Calculation

In addition to the stochastic framework implemented in MATLAB, we replicated the same calculations in an Excel spreadsheet using a deterministic projection approach obtaining the following results.

Results	MVA	BEL	d_MVA abs	d_BEL abs	d_MVA rel	d_BEL rel	BoF	d_BoF	dur_L
BASE	10000000	9577794.2	-	-	-	-	422205.799	-	5.64302804
IR_up	10000000	9575773.79	0	2020.41296	0	-0.00021095	424226.212	-2020.41296	5.64248529
IR_dw	10000000	9594004.99	0	16210.7934	0	0.001692539	405995.006	16210.79337	5.64360146
EQ	6248000	6179444.83	3752000	3398349.37	-0.3752	-0.35481545	68555.1677	353650.6313	5.69785747
PROP	9500000	9113829.9	500000	463964.303	-0.05	-0.04844167	386170.102	36035.6967	5.63971903
MORT	10000000	9580638.37	0	2844.17156	0	0.000296955	419361.627	2844.171558	5.60578915
CAT	10000000	9578343.16	0	548.958093	0	5.73157E-05	421656.841	548.9580926	5.63575668
LAPSE UP	10000000	9697430.29	0	119636.089	0	0.012490986	302569.71	119636.0891	4.02739534
LAPSE DOWN	10000000	9322844.11	0	254950.094	0	-0.02661887	677155.893	-254950.094	9.14254336
LAPSE MASS	10000000	9679368.26	0	101574.06	0	0.010605162	320631.739	101574.0603	4.2430386
EXPENSES	10000000	9582711.54	0	4917.3364	0	0.00051341	417288.463	4917.336397	5.64391654
BSCR	436617.275	-	-	-	-	-	-	-	-

Table 1.2: Deterministic Projection Results

1a Methodology

In this implementation, both equity and property components are projected deterministically using forward rates. In particular, at each time step the fund evolves according to:

$$F_t = F_{t-1} \cdot (1 + f_t)$$

where f_t is the forward rate. Moreover, the evolution is applied to the net fund, i.e. after the Regular Deduction. We adopted the convention that all cash flows occur at the end of the year and are therefore discounted using the corresponding one-year discount factor. Regarding the death cash flows we evaluated the number of deaths at time t as:

$$N_t^{death} = N_{t-1} \cdot q_t$$

where q_t are the death probabilities according to the life tables obtained on the ISTAT website. The death benefit is then computed according to the contract and discounted at time t.

The lapse cash flows are instead calculated at time t as:

$$N_t^{lapse} = N_{t-1} \cdot (1 - q_t) \cdot l$$

Lapse benefit is computed according to the contract and discounted at time t.
We then followed the same approach as in MATLAB code stressing the different scenarios.

1b Comparison with Stochastic Model

The deterministic Excel approach produces systematically lower BEL values compared to the stochastic MATLAB simulation. This difference arises from the non-linearity of the payoff, in particular for death benefits. In the stochastic framework, the fund value F_t is random, while in Excel we only consider its expected value.

In particular, for a convex payoff function such as:

$$\max(G, F_t)$$

it holds that:

$$\max(G, \mathbb{E}[F_t]) \leq \mathbb{E}[\max(G, F_t)]$$

which is a direct consequence of Jensen's inequality.

In our implementation, death and lapse are treated separately and the convexity of the death benefit implies that replacing F_t with its expectation leads to an underestimation of the liabilities.

Finally, the BSCR is based on the difference between the stressed scenarios and the base case. In Matlab, the base case is already computed over many simulated fund paths, including scenarios in which the fund performs badly. This means that the base valuation already incorporates part of the downside risk of the contract. In Excel, instead, the base case is built on a deterministic average projection, so it is less affected by adverse outcomes from the start. As a result, when a stress is applied, the additional reduction in BoF can be smaller in Matlab than in Excel. This leads to lower SCR and BSCR figures.

1c Base Case Results

1c.1 Macaulay Duration

The Macaulay duration of the liabilities is defined as:

$$\text{MacD} = \frac{\sum_{t=1}^{T} t \cdot CF_t \cdot DF_t}{\sum_{t=1}^{T} CF_t \cdot DF_t} = \frac{\sum_{t=1}^{T} t \cdot CF_t \cdot DF_t}{BEL}$$

where, as above, we consider

$$CF_t = CF_t^{death} + CF_t^{lapse} + CF_t^{exp} + CF_t^{comm}$$

In the base case, it results:

$$\boxed{\text{MacD} = 5.643028}$$

1c.2 Sources of Profit and PVFP

The main source of profit for the insurance company is the **Regular Deduction (RD)**.

The annual profit is computed as:

$$\text{Profit}_t = CF_t^{RD} - CF_t^{comm} - CF_t^{exp}$$

The Present Value of Future Profits (PVFP) is obtained as the sum of discounted annual profits:

$$PVFP = \sum_t \text{Profit}_t \cdot DF_t$$

$$\boxed{\text{PVFP} = 420{,}794.4873}$$

1c.3 Leakage Analysis

We verified the consistency relation:

$$MVA = BEL + PVFP + LEAK$$

In our case, the leakage is:

$$\boxed{\text{LEAK} = 1{,}411.3117}$$

The presence of a non-zero leakage can be explained by several aspects of the deterministic framework. First of all, the projection relies on simplified timing assumptions, since all cash flows are assumed to occur at the end of each year. Moreover, the use of expected values instead of a pathwise simulation leads to small discrepancies, especially in the presence of non-linear payoffs, as in the death benefits' one. Additional differences arise from numerical approximations in the discounting and aggregation of cash flows.

Overall, the magnitude of the leakage is relatively small compared to the total fund value, indicating a good level of consistency of the model.

1c.4 Proxy Calculation of PVFP

A practical proxy for the Present Value of Future Profits (PVFP) can be obtained by combining the first discounted profit with the duration of the profit stream.

The idea is to approximate the entire sequence of discounted profits with a simplified representation that captures both their magnitude and timing. In particular, the first discounted cash flow $CF_1 v_1$ is used as a proxy for the level of profits, while the Macaulay duration MacD summarises the average time at which these profits are received.

Under the assumption of a smooth and gradually declining profit profile, as is our case, this leads to the approximation:

$$PVFP \approx \text{MacD} \cdot CF_1 v_1$$

This approximation is not exact, but relies on the fact that when profits evolve regularly over time, the first discounted cash flow provides a reasonable estimate of the overall scale of the profit stream, while the duration captures its effective time horizon.

From an economic perspective, this approach replaces the full discounted cash flow projection with a two-parameter summary: the initial profitability and the average timing of profit emergence.

$$\boxed{\text{Proxy} = 423{,}248.1101}$$

Open Questions

1a Interest Rate Shock

What happens to the asset and liabilities when the risk-free rate increases/decreases with a parallel shift of, say, 100bps? Describe the effects for all the BEL components.

Assets

At time $t = 0$, the value of the assets does not change following a parallel shift in the interest rates. This is because the fund is composed of equity and property instruments that are marked-to-market, hence their current value is given by the observed market price F_0.

However, the interest rate level affects the future dynamics of the fund under the risk-neutral measure and therefore indirectly impacts the valuation of liabilities.

Liabilities (BEL)

An increase in interest rates leads to a decrease in the BEL, while a decrease in rates leads to an increase in the BEL. This is due to the standard discounting mechanism: higher rates imply lower discount factors and therefore lower present values of future cash flows, and vice versa.

We now analyze the impact on the main BEL components.

Lapse benefits

Lapse benefits depend on the value of the fund at the time of surrender. Since the fund evolves (in expectation, under the risk-neutral measure) at a rate linked to the risk-free rate, there is a partial offset between the increase in the fund value and the discounting effect. However, this offset is not perfect due to the volatility of the fund and the stochastic timing of lapse events.

As a result, lapse benefits exhibit a relatively low sensitivity to interest rate changes, although the effect is not negligible.

Death benefits

Death benefits are defined as the maximum between the invested premium and the fund value, introducing an element of optionality.

In the considered framework, the fund value tends to be larger than the guarantee in most scenarios, although the difference is less pronounced in the early years of the contract. Therefore, death benefits are predominantly driven by the evolution of the fund.

This leads to a reduced sensitivity to interest rates compared to deterministic cash flows, as there is a partial compensation between the expected growth of the fund and discounting. Nevertheless, the presence of the guarantee introduces a non-linear payoff, which becomes relevant in adverse scenarios for the fund.

Overall, death benefits exhibit a negative sensitivity to interest rates, but weaker than that of purely deterministic components and with non-linear effects due to the embedded optionality.

Expenses

Expenses are deterministic cash flows (indexed to inflation) and do not depend on the fund value.

Their present value is therefore entirely driven by discounting: an increase in interest rates leads to a decrease in the BEL associated with expenses, while a decrease in interest rates leads to an increase in their present value. This component is therefore the most directly and linearly sensitive to changes in interest rates.

Commissions

Commissions are proportional to the fund value and therefore indirectly depend on interest rates. As in the case of lapse benefits, there is a partial offset between the growth of the fund and discounting.

Consequently, commissions exhibit a moderate sensitivity to interest rates, lower than that of purely deterministic components, but still non-zero.

Conclusion

- assets are not affected at time $t = 0$ by an instantaneous interest rate shock;
- the BEL moves inversely with respect to interest rates;
- different BEL components show different sensitivities, mainly depending on their duration and on the presence of optional features (such as in the case of death benefits);
- this behaviour is confirmed by the numerical results obtained in the Matlab code: applying a parallel upward shock of 100 bps, the BEL decreases from 9.656×10^6 to 9.632×10^6, showing a variation that is small in magnitude but negative in sign, as expected.

Age and Gender Effects

What happens to the liabilities if the insured age increases? What if there were two model points, one male and one female?

Analyzing the different components of the liabilities, the following effects can be identified. First, with regard to death benefits, the share of exits due to death increases relative to those due to lapse, as policyholders tend to leave the portfolio earlier due to higher mortality.

With regard to the amount of death benefits, two opposite effects come into play: on the one hand, earlier payments are less discounted, increasing their present value; on the other hand, the fund has less time to grow, reducing the payoff. In the presence of the optionality max(F0, Ft), the payoff benefits from a convexity effect: if the fund Ft

grows on average at the risk-free rate, the expected value of max(F0, Ft) tends to grow more than proportionally with respect to the discount factor. In other words, as the time horizon increases, the average payoff progressively accumulates an advantage over the discount factor. Consequently, reducing the available time (as in the case of earlier deaths) attenuates this effect and the discounted death benefits are on average lower.

Lapse benefits are less numerous, as the increase in exits due to death reduces the pool of active individuals who may lapse. Their unit amount is not significantly affected, as the opposing effects of lower fund growth and lower discounting tend to offset each other.

Commissions and expenses decrease, as the base of active policyholders on which they are applied declines more rapidly over time.

This qualitative analysis might suggest an overall decrease in liabilities. However, empirical evidence obtained from our Matlab code shows the opposite result. This is attributable to the rebalancing between exits due to death and those due to lapse. In the considered model, lapse events are relatively more favorable for the insurer, as they incorporate a fixed penalty and eliminate the risk arising from the optionality in case of negative fund performance. Consequently, a reduction in lapses in favor of death benefits leads to an overall increase in liabilities.

When two model points, male and female, are introduced, the opposite effect is observed: lower female mortality implies deaths occurring later on average, a longer permanence in the portfolio, and therefore a relative increase in lapses, resulting in a lower overall BEL.

Matlab Code

The code is structured into two main parts: an initial script for data import, parameter setting, and execution of the simulations under different stress scenarios, and a core function, `COMPUTEBOF`, implementing the stochastic engine. The latter is designed in a modular way to allow flexibility in the specification of shocks and assumptions.

```
clear all;
clc;
rng(42)

rates = 'EIOPA_RFR_20251231_Term_Structures.xlsx';

t = readmatrix(rates,'Sheet','RFR_spot_no_VA','Range','B11:B60');
r = readmatrix(rates,'Sheet','RFR_spot_no_VA','Range','C11:C60');

discounts = 1./((1+r).^t);
discounts = [1;discounts];

fwd_discounts = discounts(2:end)./discounts(1:end-1);
fwd_r = log(1 + (1./fwd_discounts - 1));

up_r   = readmatrix(rates,'Sheet','Spot_NO_VA_shock_UP','Range','C11:C60');
down_r = readmatrix(rates,'Sheet','Spot_NO_VA_shock_DOWN','Range','C11:C60');

discounts_up = [1; 1./((1+up_r).^t)];
discounts_down = [1; 1./((1+down_r).^t)];

fwd_r_up = log(1 + (1./(discounts_up(2:end)./discounts_up(1:end-1)) - 1));
fwd_r_down = log(1 + (1./(discounts_down(2:end)./discounts_down(1:end-1)) - 1));

SI2024 = 'Life tables of the resident population.csv';
LT = readtable(SI2024);

age = LT{:,1};
qx = LT{:,4} / 1000;

x0 = 60;
horizon = 50;

idx = (age >= x0) & (age < x0 + horizon);
qx = qx(idx);

N0 = 100;
```

```
l = 0.15;
F0 = 1e5;

e0 = 50;
inf = 0.02;

sigma_equity = 0.2;
sigma_property = 0.1;

Nsim = 250000;

[BOF,BEL,PVFP,MacD, CI_BOF] = COMPUTEBOF(F0,qx,l,fwd_r,t,...
    Nsim,sigma_equity,sigma_property,discounts,0,inf,e0);

[BOF_up,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r_up,t,Nsim,...
    sigma_equity,sigma_property,discounts_up,0,inf,e0);

[BOF_down,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r_down,t,Nsim,...
    sigma_equity,sigma_property,discounts_down,0,inf,e0);

SCR_IR = max(max(BOF-BOF_up,0), max(BOF-BOF_down,0));

[BOF_EQ,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,1,inf,e0);

SCR_EQ = max(BOF-BOF_EQ,0);

[BOF_PR,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,2,inf,e0);

SCR_PR = max(BOF-BOF_PR,0);


qx_shock = min((1+0.15)*qx,1);
[BOF_mort,~,~,~] = COMPUTEBOF(F0,qx_shock,l,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,0,inf,e0);

SCR_mort = max(BOF-BOF_mort,0);

l_up = min(1,(1+0.5)*l);
l_down = 0.5*l;
l_mass = 0.40;

[BOF_l_up,~,~,~] = COMPUTEBOF(F0,qx,l_up,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,0,inf,e0);

[BOF_l_down,~,~,~] = COMPUTEBOF(F0,qx,l_down,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,0,inf,e0);

[BOF_l_mass,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r,t,Nsim,...
    sigma_equity,sigma_property,discounts,0,inf,e0,l_mass);
```

```
89
90 SCR_lapse = max([BOF-BOF_l_up, BOF-BOF_l_down, BOF-BOF_l_mass]);
91
92 e0_stress = e0 * 1.10;
93 inf_stress = inf + 0.01;
94
95 [BOF_exp,~,~,~] = COMPUTEBOF(F0,qx,l,fwd_r,t,Nsim,...
96     sigma_equity,sigma_property,discounts,0,inf_stress,e0_stress);
97
98 SCR_exp = max(BOF-BOF_exp,0);
99
100 SCR_life = sqrt(SCR_vec' * Corr_life * SCR_vec);
101 SCR_mkt = sqrt(SCR_vec' * Corr_mkt * SCR_vec);
102
103 BSCR = sqrt([SCR_life SCR_mkt] * ...
104              [1 0.25;0.25 1] * ...
105              [SCR_life; SCR_mkt]);
106
107
108
109 function [BOF,BEL,PVFP,MacD, CI_BOF] = COMPUTEBOF(F0, prob_death,
    prob_lapse, forward_r, t, Nsim, sigma_equity, sigma_property,
    discounts, flag, inf, e0, mass_lapse)
110
111     if nargin < 13
112         mass_lapse = prob_lapse;
113     end
114
115     if flag == 1
116         alpha = 0.39 + 0.0790370418111731;
117         shock_equity = 1 - alpha;
118         shock_property = 1;
119     elseif flag == 2
120         shock_equity = 1;
121         alpha = 0.25;
122         shock_property = 1 - alpha;
123     else
124         shock_equity = 1;
125         shock_property = 1;
126     end
127
128     F = zeros(Nsim, length(t)+1);
129     F(:,1) = F0;
130
131     F_equity = zeros(Nsim, length(t)+1);
132     F_property = zeros(Nsim, length(t)+1);
133
134     F_equity(:,1) = 0.8 * F0 * shock_equity;
135     F_property(:,1) = 0.2 * F0 * shock_property;
136
137
```

```
138     active = true(Nsim, 100);
139
140     inflation_rate = 1 + inf;
141
142     z_equity = randn(Nsim, length(t));
143     z_property = randn(Nsim, length(t));
144
145     RD = zeros(Nsim,1);
146     COMM = zeros(Nsim,1);
147     EXP = zeros(Nsim,1);
148     DEATH = zeros(Nsim,1);
149     LAPSE = zeros(Nsim,1);
150     CF_t = zeros(Nsim, length(t));
151
152     for i = 2:length(t)+1
153
154         F_equity(:,i) = F_equity(:,i-1) .* exp(forward_r(i-1) -
               0.5*sigma_equity^2 + sigma_equity*z_equity(:,i-1));
155         F_property(:,i) = F_property(:,i-1) .* exp(forward_r(i-1)
               - 0.5*sigma_property^2 + sigma_property*z_property(:,i
               -1));
156
157         num_active = sum(active, 2);
158         fund_before_rd = F_equity(:,i) + F_property(:,i);
159
160         comm_cf = 0.014 * num_active .* fund_before_rd * discounts
               (i);
161         COMM = COMM + comm_cf;
162
163         exp_cf = e0 * inflation_rate * num_active * discounts(i);
164         EXP = EXP + exp_cf;
165
166         RD = RD + 0.022 * fund_before_rd .* num_active * discounts
               (i);
167
168         F_equity(:,i) = F_equity(:,i) * (1 - 0.022);
169         F_property(:,i) = F_property(:,i) * (1 - 0.022);
170         F(:,i) = F_equity(:,i) + F_property(:,i);
171
172         death = rand(Nsim,100) < prob_death(i-1);
173         death_realized = active & death;
174         num_death = sum(death_realized, 2);
175
176         active = active & ~death;
177
178         death_benefit = max(F0 * ones(Nsim,1), F(:,i));
179         death_cf = death_benefit .* num_death * discounts(i);
180         DEATH = DEATH + death_cf;
181
182         if i == 2
183             lapse = rand(Nsim,100) < mass_lapse;
```

```
        else
            lapse = rand(Nsim,100) < prob_lapse;
        end

        lapse_realized = active & lapse;
        num_lapse = sum(lapse_realized, 2);

        active = active & ~lapse;

        lapse_cf = (F(:,i) - 20) .* num_lapse * discounts(i);
        LAPSE = LAPSE + lapse_cf;

        CF_t(:,i-1) = comm_cf + exp_cf + death_cf + lapse_cf;

        inflation_rate = inflation_rate * 1.02;
    end

    num_survivors = sum(active, 2);
    terminal_cf = F(:,end) .* num_survivors * discounts(end);
    LAPSE = LAPSE + terminal_cf;
    CF_t(:,end) = CF_t(:,end) + terminal_cf;

    if flag == 1
        MVA = (F_equity(1,1) + F_property(1,1)) * 100;
    elseif flag == 2
        MVA = (F_equity(1,1) + F_property(1,1)) * 100;
    else
        MVA = 100 * F0;
    end

    BEL_sim = COMM + EXP + LAPSE + DEATH;
    BEL = mean(BEL_sim);
    BOF = MVA - BEL;
    PVFP = mean(RD - COMM - EXP);

    mean_CF_t = mean(CF_t,1);
    MacD = sum(t' .* mean_CF_t) / BEL;

    % Confidence interval for BEL and BOF
    alpha_ci = 0.05;
    s_BEL = std(BEL_sim);
    se_BEL = s_BEL / sqrt(Nsim);
    t_crit = tinv(1 - alpha_ci/2, Nsim - 1);

    CI_BOF = MVA-[BEL + t_crit * se_BEL, BEL - t_crit * se_BEL];
end
```